Junjue Wang

Assistant Professor at The University of Tokyo

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Research Interests

My research focuses on multi-modal remote sensing and computer vision, aiming to build intelligent systems for understanding Earth observation data. My work spans three core areas: 1) Vision-Language Multi-Modal Learning: I developed the EarthVQA series for visual question answering and comprehensive understanding (EarthVQA, EarthVQANet, EarthVL, etc). 2) Large-scale Dataset Construction: I created influential benchmark datasets for domain adaptive semantic segmentation (LoveDA, LoveCS), disaster response (DisasterM3) and long-term series urban understanding (DynamicVL). 3) AI4Earth Applications: Grounding AI techniques to AI4Earth problems is interesting and challenging. I proposed CityVLM for sustainable urban development and our damage assessment solution won 1st in 'AI for Earthquake Response Challenge'.

Professional Experience

Assistant Professor 04/2026 – Present
The University of Tokyo Kashiwa, Japan
Collaborative professor: Prof. Naoto Yokoya

Project Researcher 08/2024 – 04/2026
The University of Tokyo Kashiwa, Japan
Collaborative professor: Prof. Naoto Yokoya

Education

Ph.D candidate in Photogrammetry and Remote Sensing 2019 – 2024
Wuhan University Wuhan, China
Adviser: Prof. Yanfei Zhong & Prof. Liangpei Zhang

B.E. in Surveying and Mapping Engineering 2015 – 2019
China University of Geosciences Wuhan, China
Adviser: Prof. Fang Fang & Prof. Yaqin Ye

First-Author Publications

DisasterM3: A Remote Sensing Vision-Language Dataset for Disaster Damage Assessment and Response 2025
Junjue Wang, Weihao Xuan, Heli Qi, Zhihao Liu, Kunyi Liu, Yuhan Wu, Hongruixuan Chen, Jian Song, Junshi Xia, Zhuo Zheng, Naoto Yokoya
The Thirty-Ninth Annual Conference on Neural Information Processing Systems

DynamicVL: Benchmarking Multimodal Large Language Models for Dynamic City Understanding 2025
Weihao Xuan(Co-first author), **Junjue Wang(Co-first author)**, Heli Qi, Zihang Chen, Zhuo Zheng, Yanfei Zhong, Junshi Xia, Naoto Yokoya
The Thirty-Ninth Annual Conference on Neural Information Processing Systems

CityVLM: Towards sustainable urban development via multi-view coordinated vision–language model 2025
Junjue Wang, Weihao Xuan, Heli Qi, Zihang Chen, Hongruixuan Chen, Zhuo Zheng, Junshi Xia, Yanfei Zhong, Naoto Yokoya
ISPRS Journal of Photogrammetry and Remote Sensing (ISPRS P&RS)

Cross-sensor domain adaptation for high spatial resolution urban land-cover mapping: From airborne to spaceborne imagery 2022
Junjue Wang, AilongMa, Yanfei Zhong, Zhuo. Zheng, and Liangpei Zhang
Remote Sensing of Environment (**RSE**)
ESI Highly Cited Paper (WOS citation: 47)

RSNet: The Search for Remote Sensing Deep Neural Networks in Recognition Tasks 2021
Junjue Wang, Yanfei Zhong, Zhuo Zheng, and Liangpei Zhang
IEEE Transactions on Geoscience and Remote Sensing (**TGRS**)
ESI Highly Cited Paper (WOS citation: 75)

EarthVQANet: Multi-task visual question answering for remote sensing image understanding 2024
Junjue Wang, Ailong Ma, Zihang Chen, Zhuo Zheng, Yuting Wan, Liangpei Zhang, and Yanfei Zhong
ISPRS Journal of Photogrammetry and Remote Sensing (**ISPRS**)

LoveDA: A Remote Sensing Land-Cover Dataset for Domain Adaptive Semantic Segmentation 2021
Junjue Wang, Zhuo Zheng, Ailong Ma, and Yanfei Zhong
35th Annual Conference on Neural Information Processing Systems (**NeurIPS** Datasets & Benchmarks Track)

EarthVQA: Towards Queryable Earth via Relational relational-based Remote Sensing Visual Question Answering 2024
Junjue Wang, Zhuo Zheng, Zihang Chen, Ailong Ma, Yanfei Zhong
Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI2024**)

LoveNAS: Towards Multi-Scene Land-Cover Mapping via Hierarchical Searching Adaptive Network 2024
Junjue Wang, Yanfei Zhong, Ailong Ma, Zhuo Zheng, Yuting Wan, Liangpei Zhang
ISPRS Journal of Photogrammetry and Remote Sensing (**ISPRS P&RS**)

FactSeg: Foreground Activation Driven Small Object Semantic Segmentation in Large-Scale Remote Sensing Imagery 2022
Ailong Ma, **Junjue Wang (*Corresponding Author*)**, Yanfei Zhong, Zhuo Zheng
IEEE Transactions on Geoscience and Remote Sensing (**TGRS**)

Collaborative Publications

Remote sensing meta modal representation for missing modality land cover mapping: From EarthMiss dataset to MetaRS method 2026
Yanfei Zhou, Ailong Ma, **Junjue Wang**, Zihang Chen, Yanfei Zhong
Remote Sensing of Environment (**RSE**)

Seeing is Believing, but How Much? A Comprehensive Analysis of Verbalized Calibration in Vision-Language Models 2025
Weihao Xuan, Qingcheng Zeng, Heli Qi, **Junjue Wang**, Naoto Yokoya
Conference on Empirical Methods in Natural Language Processing (**EMNLP**) 2025 Oral

BRIGHT: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response 2025
Hongruixuan Chen, Jian Song, Olivier Dietrich, Clifford Broni-Bediako, Weihao Xuan, **Junjue Wang**, Xinlei Shao, Yimin Wei, Junshi Xia, Cuiling Lan, Konrad Schindler, Naoto Yokoya
Earth System Science Data (**ESSD**)

Foundation models for remote sensing and earth observation: A survey 2025
Aoran Xiao, Weihao Xuan, **Junjue Wang**, Jiaxing Huang, Dacheng Tao, Shijian Lu, Naoto Yokoya
IEEE Geoscience and Remote Sensing Magazine (**GRSM**)

FarSeg++: Foreground-Aware Relation Network for Geospatial Object Segmentation in High Spatial Resolution Remote Sensing Imagery 2023
Zhuo Zheng, Yanfei Zhong, **Junjue Wang**, Ailong Ma, Liangpei Zhang
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)

Domain Adaptive Land-Cover Classification via Local Consistency and Global Diversity 2023
Ailong Ma, Chenyu Zheng, **Junjue Wang**, and Yanfei Zhong
IEEE Transactions on Geoscience and Remote Sensing (**TGRS**)

E2SCNet: Efficient Multiobjective Evolutionary Automatic Search for Remote Sensing Image Scene Classification Network Architecture 2022
Yuting Wan, Yanfei Zhong, Ailong Ma, **Junjue Wang**, and Liangpei Zhang
IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)

TypeFormer: Multiscale transformer with type controller for remote sensing image caption 2022
Zihang Chen, **Junjue Wang**, Ailong Ma, and Yanfei Zhong
IEEE Geoscience and Remote Sensing Letters (**GRSL**)

The Outcome of the 2022 Landslide4Sense Competition: Advanced Landslide Detection From Multisource Satellite Imagery 2022
Omid Ghorbanzadeh, Yonghao Xu, Hengwei Zhao, **Junjue Wang**, Yanfei Zhong, Dong Zhao, Qi Zang, Shuang Wang, Fahong Zhang, Yilei Shi, Xiao Xiang Zhu, Lin Bai, Weile Li, Weihang Peng, Pedram Ghamisi
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (**JSTARS**)

Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters 2021
Zhuo Zheng, Yanfei Zhong, **Junjue Wang**, Ailong Ma, and Liangpei Zhang
Remote Sensing of Environment (**RSE**)

SceneNet: Remote sensing scene classification deep learning network using multi-objective neural evolution architecture search 2021
Ailong Ma, Yuting Wan, Yanfei Zhong, **Junjue Wang**, and Liangpei Zhang
ISPRS Journal of Photogrammetry and Remote Sensing (**ISPRS P&RS**)

Cross-domain road detection based on global-local adversarial learning framework from very high resolution satellite imagery 2021
Xiaoyan Lu, Yanfei Zhong, Zhuo Zheng, **Junjue Wang**
ISPRS Journal of Photogrammetry and Remote Sensing (**ISPRS P&RS**)

Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery 2020
Zhuo Zheng, Yanfei Zhong, **Junjue Wang**, and Ailong Ma
IEEE/CVF Computer Vision and Pattern Recognition (**CVPR**)

Awards

AI for Earthquake Response Challenge (1 st place) 2025
Topic: Development of AI algorithms for global building damage assessment

Graduate Academic Innovation First Prize, Wuhan University 2022
The highest honor for graduate students of Wuhan University.

IJCAI LandSlide4Sense CDCEO workshop Challenge (1 st place) 2022
Topic: multimodal global landslide detection.

Ali Tianchi Real World Image Forgery Localization Challenge (1st place, 1/1149) 2022
Topic: certificate and file forgery detection, AI security.

IEEE GRSS Data Fusion Contest, Track: Multitemporal Semantic Change Detection 2021
Track: weakly-supervised multitemporal semantic change detection

SpaceNet 6 EarthVision workshop challenge at CVPR 2020 (Top Graduate Award, 1/411) 2020
Topic: missing-modality all-weather mapping

xView2 Challenge (4th place, 4/3500+) 2019
Topic: global building damage assessment

IEEE GRSS Data Fusion Contest (2nd place) 2019
Track: single-view semantic 3D challenge

Outstanding Graduate of China University of Geosciences 2019

Excellent Freshman Award of Wuhan University 2019
Top10 students per year

Outstanding Undergraduate Thesis Award 2019
Top 5% of undergraduates

Outstanding Undergraduate Thesis Award (Top 5% of undergraduates) 2018

9th National University GIS Skills Development Competition (1st Place) 2018

5th National University Surveying and Mapping Skills Competition (First Prize) 2018
Track: programming with C++

Academic Projects

National key research and development program (2022YFB3903404), Research backbone
"Self-learning remote sensing big data fusion real-time processing and transmission software and hardware platform", response for studying multi-modal remote sensing fusion analysis algorithms to integrate heterogeneous data (RGB, SAR, multispectral, text description, etc.), reasoning about the spatial semantic relationships between objects to assist Rural/Urban planning (water sources around the agriculture, traffic situations, etc).

National key research and development program (2017YFB0504202), Research backbone
"Real-time processing technology of land resources and ecological environment security emergency information", response for studying cross-regional and cross-sensor land-cover mapping, landslide disaster detection algorithms, such as cross-sensor domain adaptive model for global landslide detection.

National Natural Science Foundation of China Excellent Youth Fund, Research bacbone
"Hyperspectral Remote Sensing, Object Recognition, and Scene Understanding", responsible for designing a task-driven framework considering data characteristics (hyperspectral and multi-spatial resolutions, etc) to automate architecture design.

Engineering Applications

FactSeg Project: Advancing Small Objects Mapping in Large-Scale Remote Sensing Images
We held a thesis recurrence competition and the code of FactSeg was included in PaddleRS (Baidu).

LoveDA Project: Advancing Large-Scale Land Cover Mapping in Urban and Rural Areas
I organized the collection, labeling, and validation of the LoveDA dataset. I maintain Land-cover Segmentation , Urban-Rural Domain Adaptation competitions on CodaLab [314 participants]. Official code was included in TorchGeo (Microsoft), and MMCV (SenseTime).

Academic Service

Journal Reviewer
IEEE IEEE Transactions on Geoscience and Remote Sensing (TGRS)
Visual Computer
Remote Sensing
Remote Sensing Letters
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS)
International Journal of Remote Sensing (IJRS)
Section Editor for "Medical Imaging Process & Technology"
Conference Reviewer
Conference on Neural Information Processing Systems (NeurIPS) 2021-2024
IEEE Computer Vision and Pattern Recognition Conference (CVPR) 2022
IEEE International Conference on Computer Vision (ICCV) 2023

Invited Talks

High Resolution Remote Sensing Intelligent Interpretation 11/2022
Wuhan University, Geoscience Cafe #346

CapFormer: Pure Transformer for Remote Sensing Image Caption 07/2022
IEEE Geoscience and Remote Sensing Symposium, Student paper competition

Progressive Label Refinement-Based Distribution Adaptation Framework for Landslide Detection 07/2022
31st International Joint Conference On Artificial Intelligence (IJCAI), champion plan sharing

LoveDA: A Remote Sensing Land-Cover Dataset for Domain Adaptive Semantic Segmentation 12/2021
35th Annual Conference on Neural Information Processing System (NeurIPS)

Skills

Multi-modal data processing: Sentinel-2/Landsat/WorldView/QuickBird/GF-2/Capella
Remote sensing software: ArcGIS/QGIS/ENVI
Programming languages: Python/Matlab/C++/JavaScript
Deep Learning Framework: PyTorch/TensorFlow